MUHAMMAD GULAM SYARIF

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Bachelor of Computer Engineering from Universitas Brawijaya with a focus on web development, AI and Automation Engineer, robotics and the Internet of Things (IoT). Experienced in developing web applications using Laravel, Vue and React, as well as managing SQL and NoSQL databases such as MongoDB. Skilled in designing and implementing systems by applying control system methods, artificial intelligence, machine learning, and data processing. Proficient in ROS (Robot Operating System), Python, and C++ for robotics applications, and recognized as an adaptable individual with strong problem-solving abilities and the capability to work effectively in a team environment.

EDUCATION

Brawijaya Universitas – Malang, Jawa Timur — July 2022 – Jan 2026

Bachelor Degree in Computer Engineering, 3.94/4.00

EXPERIENCE

ROBOTIIK, Fakultas Ilmu Komputer, Universitas Brawijaya – Malang — Jan 2025 – Dec 2025

Head of Core Team - Humanoid Division

- Led a humanoid robotics team in designing and developing systems for the ROBOTIS OP3 robot, including task distribution, project planning, and technical decision-making.
- Monitored team progress and ensured all members met their assigned targets in alignment with project timelines.
- Implemented Zero Moment Point (ZMP) calculations and applied PID control to maintain humanoid robot balance, while integrating inverse kinematics to generate smooth and stable motion.

ROBOTIIK, Fakultas Ilmu Komputer, Universitas Brawijaya – Malang — Jan 2024 – Dec 2024

Divisi Programing

- Migrated the robotic system from ROS1 to ROS2, including rewriting and adapting control code.
- Find the robot’s balance is determined by calculating the Zero Moment Point (ZMP), using IMU data from the OpenCR controller and joint position values from each servo as input parameters.

Robotic beginner, Universitas Brawijaya – Malang — Oct 2024 – Nov 2024

Special Instructor – Electrical and Programming Division, Robin 2024

- Taught participants the fundamentals of electronics, robot wiring systems, and the application of sensors and actuators.
- Provided hands-on guidance in assembling and testing basic robotic electronic systems through practical training.

PKKMB dan Startup Academy, Universitas Brawijaya – Malang — Aug 2024 - Nov 2024

Staff Divisi Disiplin Mahasiswa (Disma)

- Ensure that all participants comply with the established rules, such as punctual attendance, appropriate clothing, and good etiquette during the event. Follow up on participants who are late or absent without a clear reason, and provide sanctions in accordance with applicable regulations.

ROBOTIIK, Fakultas Ilmu Komputer, Universitas Brawijaya – Malang — Mar 2023 – Dec 2023

Divisi Elektrikal

- Designed PCB for control buttons on the OP3 robot, performed maintenance on electrical components, and assembled robotic systems.
- Handled firmware uploading to the OpenCR controller and integrated the BNO055 IMU sensor into the robot’s sub-controller.

PKKMB dan Startup Academy, Universitas Brawijaya – Malang — Aug 2023 - Nov 2023

Staff Divisi Disiplin Mahasiswa (Disma)

- Ensure that all participants comply with the established rules, such as punctual attendance, appropriate clothing, and good etiquette during the event. Follow up on participants who are late or absent without a clear reason, and provide sanctions in accordance with applicable regulations.

PROJECT

PID-Based Zero Moment Point (ZMP) Balance Control for Humanoid Robot

- Developed a balance control system that computes the robot’s ZMP in real time and regulates it to a 0 cm setpoint using a PID controller.
- Integrated the controller with the robot’s inverse kinematics so that changes in ZMP are compensated by adjusting body posture and joint positions during both standing and walking, improving overall robot stability.

Humanoid Robot Development: Universitas Brawijaya Mascot Robot

- Currently designing and developing a humanoid robot as the official mascot for Universitas Brawijaya, with 75% of the robot's framework completed and advanced robotic control systems being integrated using ROS.

Zero Moment Point (ZMP) Implementation on Humanoid Robot OP3

- Developed a dynamic stability analysis system using ZMP and Center of Mass (CoM) calculations for the OP3 humanoid robot.

OP3 Humanoid Robot Button PCB Design

- Designed and implemented the electrical button circuit for the OP3 humanoid robot as a custom PCB using EasyEDA, including schematic capture, layout, and fabrication-ready Gerber files.

TurtleBot3 Autonomous Navigation using ROS 2 Nav2 with Waypoint Following and Obstacle Avoidance

- Developed an autonomous navigation system for TurtleBot3 using the ROS 2 Navigation Stack (Nav2) in Gazebo and RViz simulation environments, integrating LiDAR, IMU, and environmental maps to support robot localization and navigation.
- Implemented SLAM Cartographer, Adaptive Monte Carlo Localization (AMCL), Waypoint Following, Path Planning, and Obstacle Avoidance, enabling the robot to localize itself, generate optimal paths, and autonomously navigate toward target destinations while avoiding obstacles.

Wearable Device for Stress Detection Using PPG Signal

- Developed a wearable device using MAX30102 and ESP32-S3 to detect stress by extracting SDNN and AVNN features from PPG signals, then classifying stress levels with 95% accuracy using the Random Forest algorithm.

Heart Disease Detection Using Support Vector Machine Algorithm

- Developed a machine learning model using the Support Vector Machine algorithm, achieving an accuracy of 0.85 on training data.

Drug Type Classification Using k-Nearest Neighbors Algorithm

- Built a classification system for drug type determination using the k-Nearest Neighbors algorithm with K=3, achieving an accuracy of 0.82.

Fullstack Shopping Cart Application

- Developed a full-stack e-commerce shopping cart system using the MEVN stack (MongoDB, Express.js, Vue.js, Node.js) with RESTful API integration for product, cart, and order management.

Village Website Development: KKN Universitas Brawijaya

- I independently developed a Village Website using Laravel 11, Bootstrap, and Livewire.

ACHIEVEMENTSSS

- *Lulusan Terbaik pada Yudisium Fakultas Ilmu Komputer – Universitas Brawijaya, Tahun Akademik 2025/2026.*
- *Robot Sepak Bola Humanoid – Kontes Robot Indonesia (KRI) 2024.*

SKILLS

- ROS (Robot Operating System): Familiar with developing robotics applications using ROS, including sensor integration and robot control.
- PCB Design: Skilled in designing printed circuit boards (PCB) using industry standard Easy EDA software for electronic projects.
- Embedded Systems: Experienced in developing firmware and real-time applications using ESP32-S3 microcontroller, including sensor integration and wireless data transmission for wearable devices.
- Python: Proficient in Python for data analysis, signal processing, visualization, and robotics applications, including use of libraries such as Matplotlib, NumPy, and Tkinter.
- Machine Learning: Experience in developing machine learning models and algorithms for data analysis and automation.
- C++: Strong knowledge of C++ for software development, with a focus on efficiency and performance optimization.
- PHP: Expertise in PHP for server-side development, creating robust and scalable web solutions.
- HTML/CSS: Proficient in front-end web development, creating responsive and user-friendly web pages with HTML and CSS.
- Laravel: Proficient in using Laravel for web development, building dynamic and secure web applications.
- React.js: Developing interactive and responsive user interfaces using component-based architecture.
- Vue.js: Building modern front-end applications with reactive data binding and component-based design.